Hyperplane

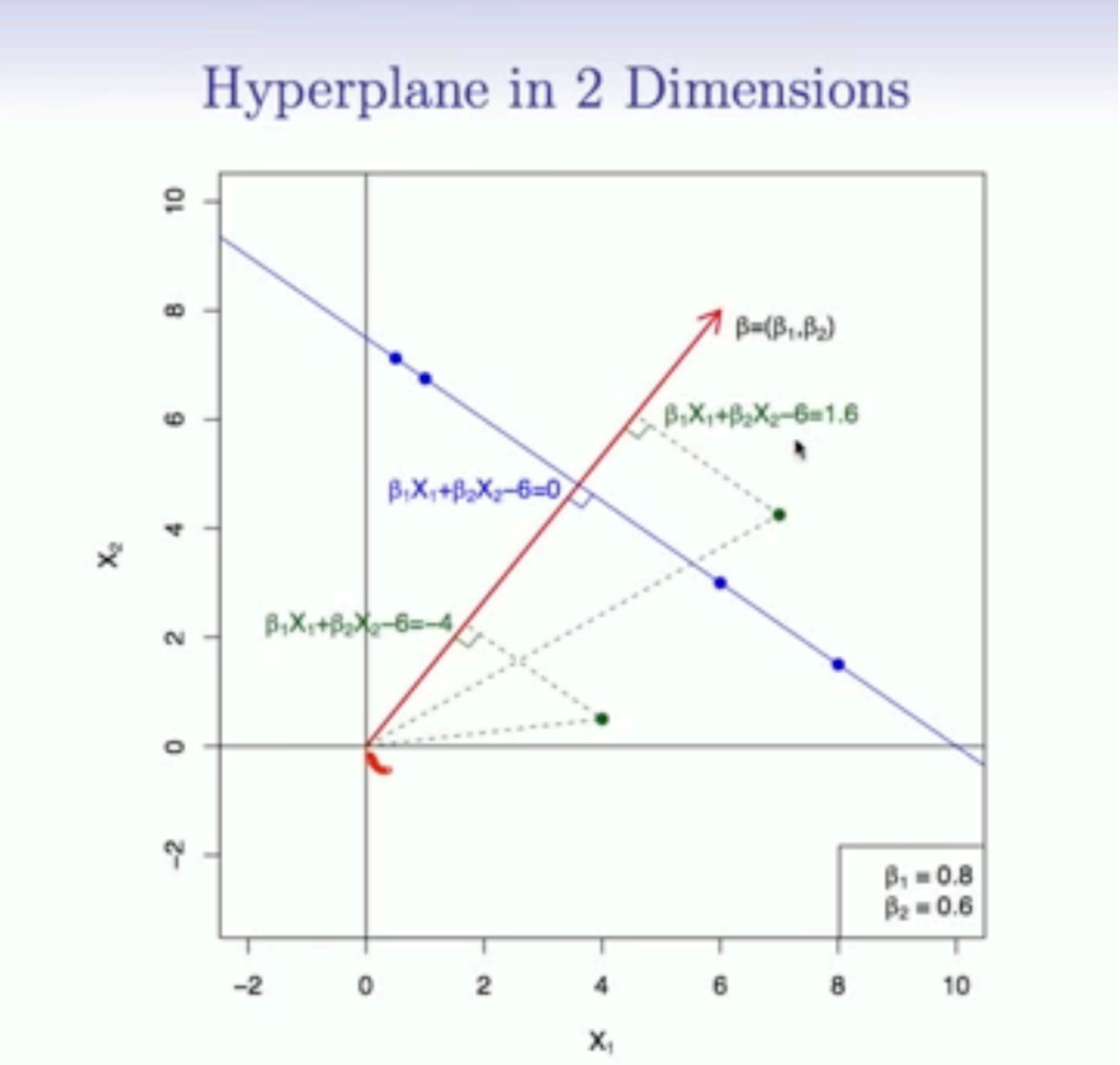
- **Hyperplane** = is a flat affine subspace of dimension p -1.
  Affine = indicates that the subspace need not pass through the origin.

In 2D, a hyperplane is a flat one-dimensional subspace = a line
in 3D, a hyperplane is a flat two-dimensional subspace = a plane
In p>3 dimension, it can be hard to visualize a hyperplane, but the notion of a (p-1)-dimensional flat subspace still applies.

In p-dimensional space, the equation for a hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

- If B0 = 0, the hyperplane goes through the origin, otherwise not.
- the vector B is called the normal vector --it points in a direction orthogonal to the surface of a hyperplane.



Separating Hyperplane

We must standardize the variable first.
as SVM treats all the variables as equal in a way, so the unit is count.